



ILLUSTRATION BY RAJ VERMA

interlocking challenges that must be addressed.

First, there exists a structural arbitrage between non-compliance and proactive investment. For many businesses, particularly in emissions-intensive sectors, the cost of inaction is often perceived as lower than the capital expenditure required for advanced sustainability technologies. This creates a lag in adoption.

Second, implementing technology often necessitates a foundational shift in organisational systems and culture. Realising the benefits of data-driven sustainability requires behavioural change, new decision-making protocols, and cross-functional buy-in.

Third, the sector faces a persistent viability gap in financing. Finally, the efficiency of digital tools in field conditions can be inconsistent.

These challenges highlight the need for robust climate governance frameworks.

## THE ROAD AHEAD

India's momentum is encouraging, but to scale climate tech from pilot to platform, three enablers must align:

### A. Policy-Tech Convergence

Sustainability regulation must evolve from periodic compliance to continuous intelligence. Integrating real-time MRV systems, AI-based scenario modelling, and digital disclosures into corporate governance frameworks is essential.

### B. Shared Climate Infra

India needs a unified, interoperable climate data stack. This infrastructure must consolidate public and private environmental dataset.

### C. Last-Mile Enablement

Technological capacity must extend to the grassroots. Local governments, urban municipalities, and village panchayats must be empow-

ered with the tools, training, and resources to act on environmental data.

India's climate challenge is immense. But so too is its capacity for innovation. We are witnessing not just a surge in cleantech entrepreneurship, but the emergence of a new national operating system—one where sustainability is embedded in infrastructure, powered by data, and driven by inclusion.

India doesn't just need climate solutions—it needs climate systems. By harnessing technology to institutionalise sustainability, the country can move from ambition to execution, from episodic action to systemic transformation.

If India can lead this transition with speed, scale, and intelligence, it will not only adapt to the global climate transition—it will define its contours. **BT**

*Views are personal*